

# Zengjun Guo

School of Urban Planning and Design,  
Peking University, China.

Email: [zengjunguo@stu.pku.edu.cn](mailto:zengjunguo@stu.pku.edu.cn)  
Website: [zengjunguo.github.io](https://zengjunguo.github.io)

## Research Interests

- **Urban Resilience & Disaster Risk Reduction:** resilience assessment of critical infrastructure systems and their interactions with human responses under compound hazards.
- **Climate Risk Modeling & Adaptation:** probabilistic simulation of climate extremes and scenario-based projections of their impacts across regions and sectors.
- **Urban Analytics & GeoAI:** large-scale spatiotemporal data and geospatial machine learning to characterize human mobility and network dynamics during disasters.

## Education

**Peking University, China** *Sep 2024–Present*

**M.Sc. in Territorial Spatial Planning**

GPA: 3.84/4.0

**Nankai University, China** *Sep 2020–Jun 2024*

**B.Mgmt. in Urban Management**

GPA: 3.85/4.0 (ranked 1st)

**University of Glasgow, UK** *Sep 2022–Jun 2023*

**Exchange Student in Urban Studies**

GPA: 18.3/22.0 (First-Class)

## Research Experience

**School of Urban Planning and Design, Shenzhen Graduate School, Peking University**

*Graduate Researcher*

*Shenzhen, China | Sep 2024–Present*

Advisor: Dr. Junqing Tang

- Generated synthetic commuting OD matrices with GLODGen and tropical cyclone tracks from CMIP6 forcing; coupled C15–TCR wind–rain modeling, asset valuation, and network assignment to assess global urban road damage and mobility disruption.
- Assessed healthcare mismatches in 52 coastal cities, annual exposure of 15.574m older adults and CNY 1.829bn total potential hospital asset losses (2015–24); projected  $2.36\times$  baseline annual losses and 19.7–32.8% climate amplification of aging-driven exposure growth under SSP5-8.5.
- Reconstructed Mangkhut’s wind–rain evolution with ER11 and GMCP and processed 2.5 billion mobile-phone records to reveal coast–inland resilience gaps and future inland vulnerability shifts.

## Key Laboratory of Earth Surface System and Human-Earth Relations (MNR)

*Research Group Member*

*Shenzhen, China | Oct 2023–Present*

Advisors: Dr. Junqing Tang and Prof. Pengjun Zhao

- Proposed a human–environment–disaster coupling theory, introduced the marginal constraint effect to define disaster risk as an endogenous constraint, and validated the theory using evidence from the 2021 Zhengzhou “7 · 20” flood.
- Mapped flood-induced shelter demand–supply mismatches across 21 cities with GHS-SMOD and developed an interactive spatial decision system for targeted emergency resource allocation.
- Refined and applied a DBN–ARIMA framework across nine Pearl River Delta cities, identifying four 2015–2021 resilience trajectories and projecting a widening Guangzhou–Shenzhen-centered core–periphery pattern through 2026.

## Selected Publications & Manuscripts

### Articles and Manuscripts

1. **Guo, Z.**, An, M., Ye, L., Zhao, P., Tang, J.\* 2026. Risk pattern and spatiotemporal evolution of healthcare-service supply in China’s coastal cities under tropical-cyclone wind disasters. **Acta Geographica Sinica** (under review). (In Chinese)
2. **Guo, Z.**, Tang, J.\*, Huang, F., Zhao, P.\* Shifting hotspots of compound tropical cyclone risk to global urban road systems in a warming climate. *In preparation.*
3. Huang, F.<sup>†</sup>, **Guo, Z.**<sup>†</sup>, Lyu, W., Zhao, P., Tang, J.\* Assessing the 15-minute city during extreme urban floods: Evidence from Shenzhen and Zhengzhou, China. *In preparation.*

4. Tang, J., **Guo, Z.**, Lyu, W., Zhao, P.\* 2026. The geographical connotation of resilient-city research: human–environment relationships under disaster-risk constraints. **Progress in Geography**, 45(4), 683–694. (In Chinese)
5. Tang, J., Lyu, W., Zhao, P.\*, Huang, F., Li, J., Fan, C., Lu, Q., Wang, J., Ye, L., **Guo, Z.** 2026. Human mobility dynamics for cost-effective flood emergency resource allocation. **Nature Sustainability**. doi:10.1038/s41893-026-01886-9.
6. Wang, J., Tang, J.\*, **Guo, Z.**, Zhao, P. 2026. Assessing spatial–temporal resilience of road transportation against extreme typhoon hazards in coastal cities: a case study in Shenzhen, China. **Journal of Transport Geography** (major revision).
7. Wang, T., Xu, T.Z.\*, Feng, H., **Guo, Z.**, Poo, M.C.-P., Tang, J., Ng, A.K.Y. 2026. Enhancing port resilience to typhoon disasters: a framework for recovery optimization and shipping network impact assessment. **Transport Policy**, 180, 104029.
8. Tang, J., Han, S., **Guo, Z.**, Ye, L., Wang, H.\* 2026. Urban resilience assessment of the Guangdong–Hong Kong–Macao Greater Bay Area based on a dynamic Bayesian network model. **Journal of Natural Disasters** (minor revision). (In Chinese)

#### **Book chapter**

9. **Guo, Z.**, Tang, J. 2026. Japan’s comprehensive efforts to mitigate damage from earthquake hazards (Ch. 11). In J. Tang, Q. Lu, P. Zhao, I. Brilakis, T. Broyd & L. Chen (Eds.), **Resilient Cities under Global Climate Change: From Principles to Applications**. Edward Elgar Publishing.

#### **Conference Presentations**

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1. “The Geographic Connotation of Resilient-City Research: An Exploration of Disaster-Risk-Constrained Human–Earth Relationships.” Oral presentation at the 13th International Conference on Population Geographies (ICPG), Nanjing, China, Jun 2026.
2. “Human–Environment Relationships under Disaster-Risk Constraints: A Preliminary Theoretical Exploration of the Geographic Connotation of Resilient-City Research.” Oral presentation at the 2026 Spring Annual Meeting of the Geographical Society of China, Chongqing, China, Apr 2026.

#### **Academic Service**

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1. Committee Member, Sub-Society for Postgraduate Students, Geographical Society of China, 2025–Present.

#### **Honors & Awards**

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1. 2026 First Prize, 10th Chengyuan Cup Planning Decision Support Model Design Competition.
2. 2025, 2026 Second Prize, Peking University Academic Science & Technology Competition.
3. 2025 Third Prize, Esri Cup China University GIS Software Development Contest.
4. 2024 National Second Prize, Undergraduate Innovation & Entrepreneurship Training Program.
5. 2024 Outstanding Undergraduate Thesis, Nankai University.
6. 2024 Outstanding Graduate, Nankai University.
7. 2024 Huadan Scholarship in Political Science, Nankai University.
8. 2021, 2023 First-Class Scholarship, Nankai University.
9. 2022 Tianjin Municipal Government Scholarship, Tianjin Municipal Education Commission.

#### **Technical Skills**

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**Programming & web development:** Python (NumPy, pandas, GeoPandas, Rasterio, xarray, scikit-learn); JavaScript; HTML/CSS.

**GIS & modeling software:** ArcGIS; GeoScene Pro; QGIS; CLIMADA; GeNIe Modeler.

**Development & typesetting:** Git/GitHub; LaTeX.